

Stochastic Simulation assignment 2A

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1 Problem description

We want to simulate waste collection in a city with $N = 3$ districts and $M = 3$ trucks. Requests arrive to each district as independent Poisson processes. Each request has a waste type that determines service-time distribution. At time 0 all trucks are idle and queues are empty.

1.1 Arrival Process

Requests arrive to district $i \in \{1, 2, 3\}$ as independent Poisson processes with rates $\lambda_1, \lambda_2, \lambda_3$. Waste type probabilities:

- Type 1 (organic): probability q_1
- Type 2 (recyclable): probability q_2
- Type 3 (general): probability $1 - q_1 - q_2$

1.2 Service-Time Distributions

- Type 1: Erlang(k_1, μ_1)
- Type 2: Erlang(k_2, μ_2)
- Type 3: Exp(μ_3)

1.3 Truck Routing and Friendliness

Each truck has a home district. After service completion, it serves home queue if non-empty. Otherwise, it inspects the other districts in a fixed cyclic priority order: truck i first checks district $(i \bmod N) + 1$, then district $((i + 1) \bmod N) + 1$, and, for each foreign district in that order: if the queue is empty, the truck proceeds to the next district; if it is non-empty, an independent Bernoulli trial with success probability p determines whether the truck starts service there, and on failure it proceeds to the next district without serving the current queue. (Concretely: truck 1 checks district 2 then 3; truck 2 checks district 3 then 1; truck 3 checks district 1 then 2.) If the truck does not start service in any foreign district (because every inspected queue was empty or every Bernoulli trial failed), it remains idle at its home district.

1.4 Rerouting Event

Rerouting is triggered by an arrival to a truck's home district: whenever a new request arrives at the truck's home district while that truck is currently serving a foreign district, evaluate the home queue length after recording this arrival; if it exceeds $K = 5$, the truck abandons its current service immediately and returns home (preemptive rerouting). The request whose service was interrupted returns to the head of that district's queue; when a truck next serves it, we draw a new service time from the same waste-type distribution (service restarts from scratch). Travel time is zero; upon rerouting the truck is immediately available to serve the home queue.

1.5 Steady-State Analysis

We will run until statistical convergence. Then we implement and justify both the regenerative method and the batch means method.

1.6 Verification Cases

- **V1 (M/M/1 reduction):** Set $N = M = 1, p = 0$, and use only Type 3 service ($\text{Exp}(\mu_3)$) with arrival rate λ_1 . The system reduces to an $M/M/1$ queue; compare simulated waiting time against

$$\mathbb{E}[W] = \frac{1}{\mu_3 - \lambda_1}, \quad \lambda_1 < \mu_3.$$

For a concrete check, use e.g. $\lambda_1 = 0.5$ and $\mu_3 = 1.0$.

- **V2 (M/M/2 reduction):** Set $N = M = 2, p = 1$, equal arrival rates $\lambda_1 = \lambda_2 = \lambda$, equal service rates $\mu_1 = \mu_2 = \mu$ (Type 3 only). Also disable rerouting for this verification (e.g. set $K = \infty$ so the threshold is never exceeded). With one truck per district ($M = N$), each truck is a dedicated server in the pooled system. By symmetry and full friendliness the system reduces to an $M/M/2$ queue; compare simulated waiting time against

$$\mathbb{E}[W_q] = \frac{C(2, a)}{2\mu - 2\lambda}, \quad a = \frac{2\lambda}{\mu}, \quad 0 < 1 < 2, \quad \mathbb{E}[W] = \mathbb{E}[W_q] + \frac{1}{\mu},$$

where $C(2, a)$ is the two-server Erlang-C delay probability (an arriving request waits) with offered load $a = 2\lambda/\mu$ in erlangs ($0 < a < 2$). In steady state this equals $P(N \geq 2)$ for the pooled $M/M/2$ system, which simplifies to

$$C(2, a) = \frac{a^2}{2 + a}$$

(equivalently $(a^2/2)/(1 + a/2)$). Sanity check: $a = 1$ gives $C(2, 1) = 1/3$, matching standard $M/M/2$ tables. For a concrete check, use e.g. $\lambda = 0.6$ and $\mu = 1.0$ (so $a = 1.2$).

1.7 Default Parameters

Unless otherwise stated, use the following parameter values:

- Arrival rates: $\lambda_1 = \lambda_2 = \lambda_3 = 0.4$
- Waste type probabilities: $q_1 = q_2 = 1/3$
- Type 1 service: Erlang($k_1 = 2, \mu_1 = 1.0$)
- Type 2 service: Erlang($k_2 = 3, \mu_2 = 1.5$)
- Type 3 service: Exp($\mu_3 = 1.0$)
- Friendliness: $p = 0.5$
- Rerouting threshold: $K = 5$

1.8 Comparison Experiment

Using the default parameters from the preceding subsection (including (μ_1, μ_2, μ_3) and waste-type probabilities), with $\lambda_1 = \lambda_2 = \lambda_3 = 0.4, q_1 = q_2 = 1/3$, we compare friendliness factors $p = 0.50$ and $p = 0.51$.

2 Metrics to compute

Waiting time. We measure waiting time as time from request arrival until service completion (total sojourn in system), consistent with the $\mathbb{E}[W]$ formulas in the verification cases above.

Queue length. Queue length counts all requests present in the district (waiting and in service). For district i , Little's law links the long-run averages as $L_i = \lambda_i \mathbb{E}[W_i]$ with λ_i the arrival rate to that district. For overall city-wide aggregates, use $\lambda_{tot} = \lambda_1 + \lambda_2 + \lambda_3$ with the overall mean sojourn time

$$\mathbb{E}[W_{tot}] = \sum_{i=1}^3 \frac{\lambda_i}{\lambda_{tot}} \mathbb{E}[W_i]$$

so that $L_{tot} = \sum_i L_i = \lambda_{tot} \mathbb{E}[W_{tot}]$ in the combined system.

Truck utilisation. For each truck, we report the long-run fraction of time that truck is busy serving any request (home or foreign district).

- Expected steady-state waiting time per district (95% CI)
- Expected steady-state queue length per district(95% CI)
- Truck utilisation per truck (95% CI)
- Rerouting rate (events per unit time)

3 State variables

The state variables are:

- Queue list per district Q_i , with stored arrival time and waste type
- Status of a truck j , S_j is either idle or busy
- The current district truck j is doing service, $D_j = i$
- Home district of a truck H_j
- On which request the truck is working, with start end completion time

4 Events

In this model we assume that whenever a request arrives and the hometruck is busy (but the queue length is $K \leq 5$), if applicable, the other idle truck(s) are alerted and called upon.

- Arrival of a request in district i
 - Request joins the system with a waste type (service time), and arrival time is stored
 - Place request in queue of district i , Q_i
 - The next arrival of a request for district i is scheduled
 - If the home truck, $j = i$, is idle, start service
 - Else, for each idle truck, trigger the truck decision procedure
 - If $Q_i > K$ schedule a rerouting for home truck from district i
- Truck decision procedure
 - Check if the home queue, Q_j is non-empty, then start service at home district
 - For each district k starting at $(i \bmod N) + 1$ if Q_k is non-empty. Then do a Bernoulli trail with probability p to start service.
 - If no service is started remain idle
- Start service truck j in district i
 - Remove first request from Q_i
 - Set status truck j , S_j to busy, and set truck j doing service at district i , $D_j = i$
 - Schedule end service
- Completion of a request by truck j
 - Set status truck j to idle
 - Trigger truck decision procedure for truck j
 - Number of completed requests increases
 - Store waiting time (=completion time-arrival time)
- Rerouting of truck j
 - Return current request to the front of its original queue Q_i
 - Cancel the scheduled completion event
 - Increase rerouting counter
 - Schedule start service immediately with first request at home queue
 - Trigger decision procedure for all idle trucks

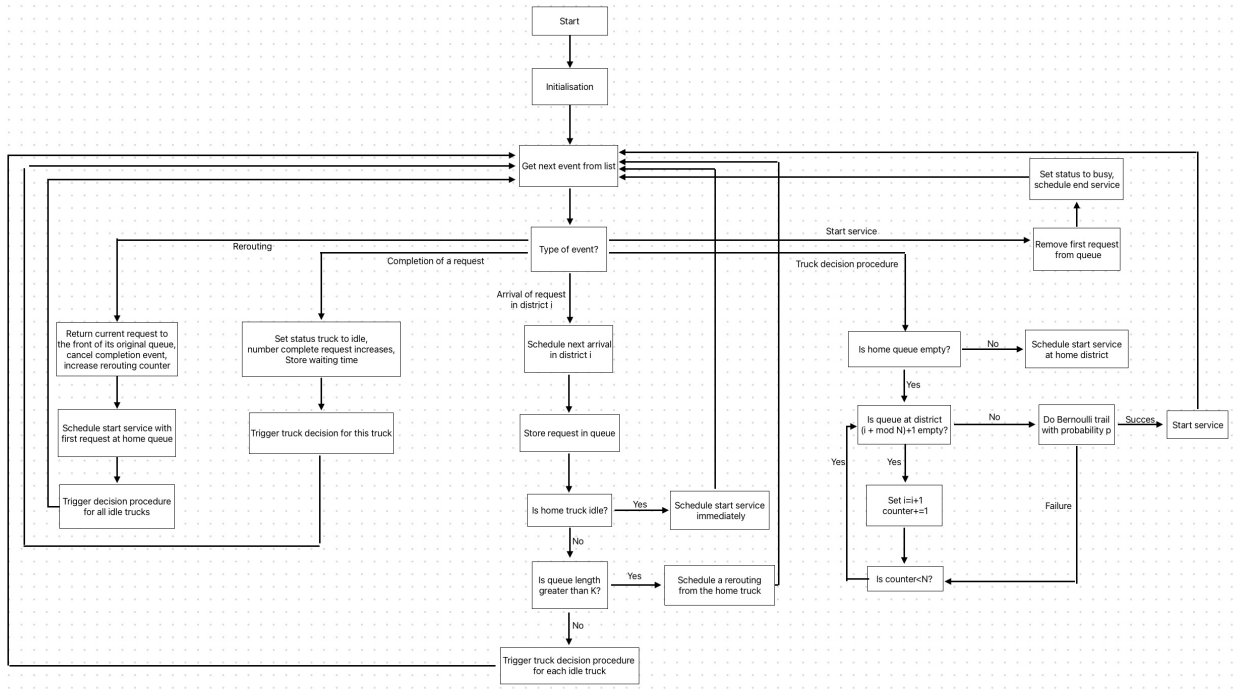


Figure 1: Flowchart

5 Flowchart

6 Justification of steady-state analysis

6.1 Regenerative method

For the regenerative method, we use the fact that when the system is completely empty it starts over, and that is independent of the previous period. So can we define possibly multiple periods (if system gets empty) and then find steady state parameters. To implement this method in the simulation we added a function `check_regeneration` which after every event checks whether the system is empty. So it checks whether all queues are empty, and all trucks are idle. If that is the case we can determine the length of the cycle, and append all statical counters to a history list to keep all the information. Then reset everthing and start over. When we have reached the end of the simulation, we can then calculate the means and variances to determine a confidence interval.

6.2 Batch means method

For the batch means method we rely on the idea that after some time we will end up in a steady state. Therefore we will first have to go through a warm-up period of which we neglect the values of the variables (and statical counters). In our simulation we determine the warm-up length as in the lecture. We have the function `determine_warmup` which overrides the execute of the service completion temporary. Then it calculates for a given number of runs N_W , W_{nk} which is the k 'th observation in the n -th run of given length K_W (which is large). Then for every run we compute the point estimator $\bar{W}_{\cdot k} = \frac{1}{N_W} \sum_{n=1}^{N_W} W_{nk}$. After that we determine the end of the warm-up period D by iterating over all values between 100 and $0.5K$ such that $\left| \frac{\frac{1}{2D} \sum_{k=1}^{2D} \bar{W}_{\cdot k}}{\frac{1}{D} \sum_{k=1}^D \bar{W}_{\cdot k}} - 1 \right| \leq 0.05$. Once we have found that we can run our actual simulation starting from time D and split it up in batch lengths of size $b = 4D$. We also need to determine the number of batches we run. We did this by calculating the relative precision by halfwidth/mean. When that is smaller then a certain δ we accepted it. To find a good amount of number of batches we just ran it with a certain number and then looked whether the condition holds.

7 Answers to the verification cases

7.1 M/M/1 reduction

Running the simulation with $N = M = 1$, $p = 0$, $q_1 = q_2 = 0$, $q_3 = 1$, $\lambda_1 = 0.5$, and $\mu_3 = 1.0$ yields the result: District 0 : 2.0432(95% CI:[1.9520, 2.1344]), for the regenerative method.

District 0 : 2.0369 (95% CI: [1.9515, 2.1222]), Relative precision: 0.0419. For the batch means method. Which is in line with the expected value for the waiting time $E[W] = \frac{1}{\mu_3 - \lambda_1} = \frac{1}{1.0 - 0.5} = \frac{1}{0.5} = 2$.

7.2 M/M/2 reduction

Running the simulation with $N = M = 2$, $p = 1.0$, $q_1 = q_2 = 0$, $q_3 = 1$, $\lambda_1 = 0.6$, $\mu_3 = 1.0$ and $K = \infty$ yields the result:

District 0 : 1.5835(95% CI:[1.5302, 1.6369]),

District 1 : 1.5532(95% CI:[1.5059, 1.6006]), for the regenerative method.

District 0 : 1.5835 (95% CI: [1.5240, 1.6430]) Relative precision: 0.0376

District 1: 1.5768 (95% CI: [1.5149, 1.6386]) Relative precision: 0.0392. For the batch means method.

Which is again in line with the expectation, since $E[W] = E[W_q] + \frac{1}{\mu} = \frac{c(2,a)}{2\mu-2\lambda} + \frac{1}{1} = \frac{c(2,1.2)}{2 \cdot 1.0 - 2 \cdot 0.6} + 1 = \frac{0.45}{0.8} + 1 = 0.5625 + 1 = 1.5625$, where $c(2, 1.2) = \frac{(1.2)^2}{2+1.2} = 0.45$.

8 Results of the comparison experiment

In the tables below results of the simulations are given for the default parameters and $p = 0.50$, and $p = 0.51$.

District	1	2	3
$p = 0.50$	2.5404 (95% CI: [2.4761, 2.6048])	2.5190 (95% CI: [2.4562, 2.5819])	2.5144 (95% CI: [2.4551, 2.5738])
$p = 0.51$	2.5389(95% CI:[2.4566, 2.6211])	2.5552(95% CI:[2.4672, 2.6432])	2.5060(95% CI:[2.4056, 2.6064])

Table 1: Expected Steady-State Waiting Time for the batch means method

District	1	2	3
$p = 0.50$	2.5455(95% CI:[2.4984, 2.5925])	2.5161(95% CI:[2.4724, 2.5599])	2.4947(95% CI:[2.4486, 2.5408])
$p = 0.51$	2.5719(95% CI:[2.5215, 2.6222])	2.5595(95% CI:[2.5110, 2.6079])	2.5528(95% CI:[2.5016, 2.6039])

Table 2: Expected Steady-State Waiting Time for the regenerative method

The friendliness factor only changes slight a bit, therefore it is not unexpected that for the waiting time, there are also no major changes. The confidence intervals almost completely overlap, and the values are really close to eachother.

9 Other computed Metrics

9.1 Batch means method

District	1	2	3
Queue length*	1.0356 (95% CI: [0.9992, 1.0720])	1.0250 (95% CI:[0.9898, 1.0602])	1.0176 (95% CI:[0.9818, 1.0535])
Truck util**	0.7177(95% CI:[0.7106, 0.7248])	0.6681(95% CI:[0.6607, 0.6755])	0.6082(95% CI:[0.5982, 0.6182])

Table 3: Results for metrics we computed per district/truck with the Batch means method, where the relative condition precision holds for all of the above results. *: Expected steady-state queue length, **: Truck utilisation per truck

Rerouting rate: 0.000217

9.2 Regenerative method

District	1	2	3
Queue length*	1.0075 (95% CI: [0.9847, 1.0304])	1.0035 (95% CI:[0.9806, 1.0263])	0.9931 (95% CI:[0.9694, 1.0168])
Truck util**	0.7180(95% CI:[0.7132, 0.7228])	0.6656(95% CI:[0.6599, 0.6713])	0.6063(95% CI:[0.5997, 0.6129])

Table 4: Results for metrics we computed per district/truck with the Regenerative method. *: Expected steady-state queue length, **: Truck utilisation per truck

Rerouting rate: 0.000134